SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1703**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 5

Duration : 1 Hour
Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

I K.Pujitha (192524149) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K.Pujitha

Annexure A

Analytical Problem Solving

1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.

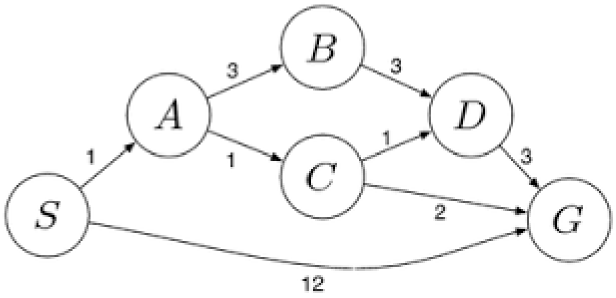
2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.
   i. What are the percept's for this agent?
   ii. Characterize the operating environment.
   iii. What are the actions the agent can take?
   iv. How can one evaluate the performance of the agent?
   v. What sort of agent architecture do you think is most suitable for this agent? Why?

3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.

4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.

5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path from the starting warehouse **S** to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.

   The delivery network is represented as a weighted graph:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem

1. <u>Water Jug Problem</u>

Problem

Two jugs are available:

- Jug A = 4 gallons
- Jug B = 3 gallons

Find exactly 2 gallons in the 4-gallon jug.

Initial State

(0,0)

Goal State

(2,0)

Solution Steps

Step	Action	State (4-Gallon, 3-Gallon)
1	Fill 3-gallon jug	(0,3)
2	Pour into 4-gallon jug	(3,0)
3	Fill 3-gallon jug again	(3,3)
4	Pour into 4-gallon jug until full	(4,2)
5	Empty 4-gallon jug	(0,2)
6	Pour remaining 2 gallons into 4-gallon jug	(2,0)

Goal Achieved

The 4-gallon jug contains exactly 2 gallons**.**

2. **Mars Rover Agent**

i. Percepts

The rover receives information from:

- Cameras
- Temperature sensor
- Soil sensor
- Rock sensor
- Distance sensor
- GPS
- Battery level

ii. Operating Environment

- Partially Observable
- Deterministic
- Sequential
- Dynamic
- Continuous
- Single Agent

iii. Actions

- Move forward/backward
- Turn left/right
- Capture images
- Collect soil samples
- Analyze rocks
- Send data to Earth
- Avoid obstacles
- Recharge (solar panels)

iv. Performance Measure

The rover is evaluated based on:

- Number of successful samples collected

- Accurate scientific analysis
- Safe navigation
- Battery efficiency
- Successful communication with Earth

v. Suitable Agent Architecture

Utility-Based Agent

Reason

A utility-based agent selects the best action by considering:

- Safety
- Energy consumption
- Scientific value
- Distance
- Time

Hence it is most suitable for Mars Rover exploration.

3. 8-Queens Problem

Problem Components

Initial State

Empty chessboard.

State Space

All possible arrangements of eight queens.

Operators

Place one queen in a column.

Goal Test

No two queens attack each other.

That means:

- No same row
- No same column
- No same diagonal

Path Cost

Each queen placement has equal cost.

Search Strategy

Backtracking Search (Depth First Search)

Steps

1. Place first queen.
2. Move to next column.
3. Check safety.
4. If safe, place queen.
5. Otherwise backtrack.
6. Continue until all 8 queens are placed.

Result

A valid arrangement where no queen attacks another queen.

4. <u>OLA Cab Booking Agent</u>

Type of Agent

Goal-Based Agent

Reason

The customer's goal is to reach the destination using the most suitable cab.

The agent considers:

- Pickup location
- Destination
- Fare
- Cab availability
- Travel time
- Customer preference

Then selects the best cab.

Pseudocode

```
START

Input Pickup Location
Input Destination

Display available cab types

User selects cab

Check availability

IF cab available THEN
    Calculate fare
    Estimate arrival time
    Confirm booking
ELSE
    Display "Cab Not Available"
ENDIF
```

END

Advantages

- Easy booking
- Chooses suitable cab
- Saves time
- Improves customer satisfaction

5. **Uniform Cost Search (UCS)**

Given Graph

Edges:

- S → A = 1
- S → G = 12
- A → B = 3
- A → C = 1
- B → D = 3
- C → D = 1
- C → G = 2
- D → G = 3

UCS Expansion

Start at S

Open List:
S(0)

Expand S

- A = 1
- G = 12

Choose lowest cost:
A(1)

Expand A

- B = 4
- C = 2

Choose lowest:
C(2)

Expand C

- D = 3
- G = 4

Choose lowest:
D(3)

Expand D

- G = 6

Open list:

- G = 4
- B = 4
- G = 12

Choose G(4)

Goal reached.

Shortest Path

S → A → C → G

Total Cost

= 1 + 1 + 2

= 4

UCS Search Table

Step	Expanded Node	Cost
1	S	0
2	A	1
3	C	2
4	D	3
5	G	4

Final Answer

- Shortest Path: S → A → C → G
- Minimum Cost: 4